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Yale Investments Office: November 2020

On a late November afternoon in 2020, David Swensen, chief investment officer of Yale University, looked out his window at the autumn leaves from his home in Killingworth, Connecticut. He reminisced about the 2019 Harvard-Yale football game, which had taken place almost precisely a year earlier. Just before the start of The Game, Yale's Athletic Director Vicky Chun surprised Swensen and his wife Meghan McMahon by informing them that they would be responsible for the official coin toss. The ensuing football contest did not disappoint, as Yale made an impressive 17-point comeback in the fourth quarter and won in double overtime. Swensen looked back at The Game as one of the highlights of his 45-year association with Yale as a student, teacher, and investor.

That memory, of a simple heads-or-tails proposition, of the unbridled joy of a victorious hometown crowd, seemed a universe away now. Swensen turned his thoughts to the pandemic, which had brought death, illness, and financial distress to so many. With regard to the endowment, he pondered the work that he and his colleagues had undertaken in the spring to understand the path of the pandemic, its economic implications, and its effects on markets. With the benefit of hindsight, Investments Office base case projections had been overly pessimistic. Had Swensen been wrong, or just early?

Since his arrival in 1985, Swensen had steered Yale's endowment—now the second-largest university endowment in the world—from $1 billion to an all-time high of $31.2 billion at the end of the 2020 fiscal year. For the 35 years ending June 30, 2020, the endowment compounded at an outstanding 13.1% per annum. Yale's student body had grown modestly during this period, its faculty significantly, and the building space considerably, culminating with the opening of two new residential colleges—its first new colleges in more than half-a-century—in 2017.

Yale pioneered an unconventional approach to managing its endowment. Rather than keeping a substantial share of its assets in domestic equities and bonds, as was the norm in the mid-1980s, Yale made significant investments in less efficient markets such as leveraged buyouts, venture capital, real estate, natural resources (timber, farmland, oil, and gas), and absolute-return investments (hedge funds). Until the 2008 financial crisis, Yale's approach had generated a string of nearly uninterrupted enviable returns and led a number of other organizations to attempt to emulate the "Yale Model."

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Swensen and his staff, while proud of their record, had long cautioned that it should not be blindly adopted.

Though the financial crisis led to a nearly 25% loss in endowment value, Swensen himself was philosophical about such fluctuations. "I'd rather have the best long-term record," he observed, "than the best one-year record. This is a long-term game." And indeed, Yale's long-term record was second to none: over the three decades ending June 30, 2020, Yale's 12.4% annual returns far exceeded that of the typical endowment (8.7%), resulting in $33.8 billion of added value that manifested itself in both higher spending distributions and higher endowment values. Compared to a classic 60% stock-40% bond portfolio, Yale added $36.6 billion over the same time period.

But at the same time, Yale's recent history and world events did pose some questions. How would the COVID-19 pandemic influence endowment growth and spending over the next few years? As more endowments and other limited partners (LPs) followed Yale's model, how would Yale maintain its competitive advantage? Finally, how should Yale respond to the changes facing the leveraged buyout and venture capital sectors, which had historically played an integral role in its portfolio?

Background[1]

Ten Connecticut clergymen established Yale in 1701. Over its first century, the college relied on the generosity of the Connecticut General Assembly, which provided more than half of its funding. The 1818 disestablishment of Congregationalism as Connecticut's state religion triggered the creation of a formal endowment for Yale. By the century's end, the endowment had reached $5 million.

The growth of the endowment accelerated during the first three decades of the twentieth century, due both to several enormous bequests and to aggressive investments in equities, which represented well over half the endowment's portfolio during the Roaring Twenties. In 1930, equities were 42% of the Yale endowment; the average university had only 11.5%.[2] Yale avoided severe erosion of its endowment during the Great Depression in the 1930s, however, because many recent bequests were kept in cash or Treasuries rather than being invested in equities.

In the late 1930s, Treasurer Laurence Tighe decided that the share of equities in Yale's portfolio should be dramatically reduced. Tighe argued that higher taxes were likely to expropriate any corporate profits that equity holders would otherwise receive even if a recovery did occur. He concluded that bonds would consequently perform better than stocks. His decision, which stipulated that at least $2 would be held in fixed-income instruments for every $1 of equity, set the template for Yale's asset allocation over the next three decades. These policies seemed very prudent in the late 1930s and 1940s. But, unfortunately, they were less well suited for the bull market of the 1950s and 1960s. In the mid and late 1960s, in response, the endowment's trustees decided on two substantial policy shifts.

First, the trustees decided to increase substantially the university's exposure to equity investments. In this decision, they were influenced by the findings of a blue-ribbon task force that argued most university endowments had taken too conservative an approach: "It is our conclusion that past thinking by many endowment managers has been overly influenced by fear of another major crash. Although nobody can ever be certain what the future may bring, we do not think that a long-term policy founded on such fear can survive dispassionate analysis."[3]

Second, Yale decided to contract out much of the portfolio management function to an external adviser, which it helped to found. Boston-based Endowment Management and Research Corporation (EM&R) boasted principals who were successful growth-stock investors recruited from other Boston

money management firms. EM&R would function as a quasi-independent external firm, free to recruit additional clients. Yale would be its largest client, with priority over other clients.

EM&R never fulfilled its high expectations. Like other universities, Yale saw its endowment's value plummet in the ensuing years due to a bear market, accelerating inflation, and operating deficits. Between June 30, 1969 and June 30, 1979, the inflation-adjusted value of Yale's endowment declined by more than 40%. Although this performance was not unusual relative to other endowments, it severely strained the university's financial fabric. Yale terminated its relationship with EM&R in 1979 and began using a variety of external advisers in its evolving asset management framework.

David Swensen and the Investments Office

In 1985, David Swensen was hired to head the Investments Office. William Brainard, Yale's provost at the time, and James Tobin persuaded their former student—Swensen had earned his PhD in economics at Yale in 1980—to leave his post at Lehman Brothers. The position offered not only the opportunity to help Yale, but the possibility of some teaching at Yale College as well.

Since then, Swensen had built the capabilities of the Yale Investments Office. Most important, he recruited and developed a high-quality internal staff. Dean Takahashi, whom Swensen had known as a Yale student, joined the Investments Office, where he became Swensen's primary lieutenant for more than three decades until 2019, when he stepped away to found the Yale Carbon Containment Lab.[4] Many of the staff were hired directly out of Yale College or one of its graduate schools, and some had unusual backgrounds, such as the classics student who became a financial analyst. As of June 30, 2020, 16 of the office's 23 investment professionals had come from Yale. "To be interested in this job, it helps a lot if you feel a real connection to Yale and want to do something good for the University," said R. Alexander Hetherington, a Director and a 2006 Yale graduate.[5] Swensen encouraged his staff to be active members of the larger Yale community, and he had chosen his office's near-campus location to signal that the Investments Office was an integral part of the university and its financial management function.

Swensen was proud of the Investments Office's history of supplying leaders to other institutions. Office alumni led investing activities at Bowdoin, the Packard Foundation, the Metropolitan Museum of Art, MIT, Princeton, Stanford, Wesleyan, and the University of Pennsylvania. The full list ran to 14 institutions, of which 9 were headed by women.

Swensen defined the role of the Investments Office broadly. Reporting to the president and to an Investment Committee (described below), the Investments Office had overall responsibility for endowment matters. While most of its day-to-day activities involved evaluating, selecting, monitoring, and overseeing external investment advisers, it also played a critical role in the policymaking process. For example, it was responsible for recommendations on both the investment policy and the spending policy for the endowment—that is, in broad terms, how the money should be invested and how much of it could be spent in any given year.

The Investment Committee was composed of influential and knowledgeable Yale alumni, a number of whom were active in asset management. The committee as a whole functioned as an involved board, meeting quarterly and providing advice and counsel to the investment staff, and ultimately approving the various investment strategies. In addition, Swensen often consulted with individual members of the Investment Committee on issues within their areas of expertise. This helped guide the thinking and recommendations of the Investments Office on various important issues, and fostered an atmosphere of advice and support within which the Investments Office could take quite different and sometimes

unconventional stances. This support was critical during the crisis of 2008, when rather than panicking and selling illiquid partnership interests at steep discounts, as some of its peers did, Yale actually bought some on the secondary market. "In hindsight," commented Swensen, "we should have bought more."

Investment Philosophy

Yale's investment philosophy was perhaps the most important factor that drove the university to reject a conventional approach to portfolio construction. Swensen was fond of quoting John Maynard Keynes's maxim "Worldly wisdom teaches us that it is better for reputation to fail conventionally than to succeed unconventionally."[6] Nonetheless, Swensen was willing to take "the risk of being different" when it seemed appropriate and potentially rewarding. By not following the crowd, Yale could develop its investment philosophy from first principles, which are summarized below.

First, Swensen strongly believed in **equities**, whether publicly traded or private. He pointed out that equities are a claim on a real stream of income, as opposed to a contractual sequence of nominal cash flows (such as bonds). Since the bulk of a university's outlays were devoted to salaries, inflation could place tremendous pressure on its finances. Not only did bonds have low expected returns relative to more equity-like assets, but they often performed poorly during periods of rising or highly uncertain inflation. To demonstrate convincingly why he believed in the long-run advantages of equity investing, Swensen referred to the actual cumulative long-run returns over past decades. An original $1 investment in December 1925 in large-company U.S. stocks would be worth $9,244 by the end of 2019, and one in small-company stocks $39,381; a comparable investment in U.S. Treasury bonds would be worth $159, and one in Treasury bills $22.[7] With Yale's current target asset allocations, more than 90% of the endowment's assets were expected to produce equity-like returns.

A second principle was **diversification**. In general, Yale believed that risk could be more effectively reduced by limiting aggregate exposure to any single asset class, rather than by attempting to time markets. While Swensen and his staff had their own informed views of the economy and markets, they believed that those views were usually reflected in market prices. They thus tended to avoid trying to time short-run market fluctuations and would overweight or underweight an asset class only if a persuasive case could be made that market prices were measurably mis-valued for understandable reasons.[8]

A third principle was to seek opportunities in **less efficient markets**. Swensen noted that from 2005 to 2019, the difference in performance between U.S. fixed-income managers from the median to the fifth percentile (of their performance universe) was only 1.4% per year, and the comparable difference in performance for U.S. large-capitalization value stock portfolio managers was 1.9% per annum. This gap widened in less liquid assets: 6% for hedge funds, 15.5% for global real estate, 21.2% for global buyout funds, and 29.4% for global venture capital.[9] This suggested that incremental returns could increase by selecting superior managers in nonpublic markets characterized by incomplete information and illiquidity, and that is exactly what Yale endeavored to do. As a result, the 2021 target allocation for public stocks, bonds, and cash was only 21.5% of Yale's portfolio.

Fourth, Swensen believed strongly in using **outside managers** for all but the most routine or indexed investments. These external investment advisers were given considerable autonomy to implement their strategies as they saw fit, with relatively little interference from Yale. The managers were chosen carefully after a lengthy and probing analysis of their abilities, comparative advantages, performance records, and reputations. The Investments Office staff was responsible for developing close and mutually beneficial relationships with each of these external managers. The staff prided

themselves on knowing their managers well, listening carefully to their ongoing advice, and helping to guide them, if and when appropriate, on policy matters. Occasionally, the Investments Office effectively "put a team in business" by becoming a new manager's first client. Managers often considered Yale one of their most important clients.

Finally, the Yale philosophy focused critically on the explicit and implicit **incentives** facing outside managers. In Swensen's view, most of the asset management business had poorly aligned incentives built into typical client-manager relationships. For instance, managers typically prospered if their assets under management grew large, not necessarily if they performed well for their clients. The Investments Office tried to structure innovative relationships and fee structures with their external managers to align the managers' interests as closely as possible with those of Yale.

Recent Asset Allocation and Performance Results

Yale's Investment Committee reviewed its endowment portfolio annually to decide on target allocations to the various asset classes. The actual allocations in recent years are shown in **Exhibit 1**, as well as the current (2021) target allocations. The comparable asset allocations for several groups of university endowments are shown in **Exhibit 2** and **Exhibit 3**. Private equity allocations for large institutions (including both pension funds and endowments) are shown in **Exhibit 4,** and **Exhibit 5** shows inflows into private equity by sub-asset class.

As a part of its planning process, the Investments Office performed a mean-variance analysis of the expected returns and risks from its current allocation and compared them with those of past Yale allocations and the current mean allocation of other universities. These computations relied on specific assumptions about the expected returns, volatilities, and correlations among asset classes. The expected real return of each asset class was built upon: (a) the real risk-free rate; (b) a market risk premium for that asset class; and (c) the excess return that Yale might expect to earn.

Mean-variance analysis posed several issues. First, because relationships could change dramatically over time, the Investments Office did not just rely mechanically on historical data, but instead modified the historical numbers based on its own experience. Second, the Investments Office imposed limits on the amount that could be invested in each asset class. If it did not, the optimization program would instruct Yale to hold no domestic equities (or even to short-sell this asset class) and instead invest in the more illiquid or higher expected excess return alternatives. This result followed naturally from the assumptions of the model: for instance, emerging foreign equities were projected to have nearly twice the real return of U.S. equities (11.0% vs. 6.0%), albeit with a higher standard deviation (23.0% vs. 17.0%). The imposition of these constraints reflected the need of the university to diversify its holdings and limit illiquidity. **Exhibit 6** shows the comparative mean-variance analysis results.

In addition, the Investments Office examined the long-run implications of its allocation for the downside risk to the endowment value and the short-term implications for stability in spending. Staff estimated the probability that the inflation-adjusted value of the endowment would fall by more than 50% over the next 50 years and the probability that the available endowment spending would fall by more than 10% (adjusted for inflation) over a two-year period. Employing Monte Carlo simulations, the Investments Office compiled thousands of possible random outcomes drawn from the assumed distribution of returns produced by the mean-variance analysis. This downside-risk analysis suggested that, based on Yale's current actual asset allocation, the probability of a 50% purchasing power decrease over a 50-year horizon was 2% and that of a 10% spending drop within any two-year period was 5%.

Scenario analysis was particularly valuable for planning during abnormal events, such as the COVID-19 crisis. In its scenario planning, the Investments Office first considered multiple trajectories of viral spread and vaccine development. Then it determined the likely economic implications of each public health outcome, including government stimulus response. Next, the team layered in how capital markets might behave and projected how Yale's asset classes would perform in each scenario over the next ten years. The scenarios were informed by input from Yale professors in economics, financial markets, public health, and biology. The four most likely scenarios developed were:

- A "base case," characterized by periodic economic lock-downs and partial re-openings until a vaccine or treatment was made widely available in 2022. In this scenario, there would be elevated unemployment and economic damage, but government stimulus would prevent a depression and financial recovery would be achieved by 2024.
- A "depression case" that reflected a long vaccine development period with severe economic and financial consequences throughout the decade.
- An "inflation case," in which the vaccine was ready by 2022 but stimulus and shortages caused high inflation for several years.
- An "optimistic case" of a vaccine ready in 2021 with a full economic recovery in 2022.

Scenario planning could not perfectly predict the future, but it helped Yale prepare for how to manage endowment growth, asset allocation, and university spending across multiple likely outcomes as seen in **Exhibit 7**. In particular, the analysis highlighted the need to be prepared for a significant downside case. As a result, the Investments Office modified the endowment's asset allocation targets and planned some redemptions from marketable managers. More importantly, Yale decided to issue taxable debt and use the proceeds to cover operating costs, which permitted the endowment to delay about $1.0 billion of planned spending distributions. Alex Banker HBS '91, Director of Risk Management and Capital Markets Activities, noted, "Yale took advantage of historically low interest rates to bolster the liquidity position of the University." Of the $1.5 billion in debt, $500 million had a maturity of 5 years at a spread to the corresponding U.S. Treasury bond of 0.55% and a yield of 0.87%, $500 million had a maturity of 10 years at a spread of 0.80% and a yield of 1.48%, and $500 million had a maturity of 30 years at a spread of 0.93% and a yield of 2.40%. At the time of issuance, the spread to the 30-year Treasury was the lowest to date among AAA peers.

Underpinning the scenario analysis was the fortunate fact that Yale entered 2020 after decades of extraordinary performance. Some of this could be ascribed to diversification, with each asset class making positive contributions even while occasionally experiencing "bouts of short-term underperformance."[10] Over the prior two decades (to June 30, 2020), Yale's annualized returns of 9.9% exceeded the returns of domestic stocks (6.2%) and bonds (5.1%).[11] These results bested the performance of similar sized endowments (more than $1 billion) by 3.4% per year.[12] This performance also exceeded Yale's large peers (8.4%) and universities as a whole (5.8%).

Exhibit 8 shows the endowment's recent performance compared with that of other universities; **Exhibit 9** includes a more detailed breakdown of Yale's returns by asset class. Not only had the average return been high, but the endowment produced consistent results: Yale had experienced only two fiscal years with negative returns (1988, which included the market crash in October 1987, and 2009, which included the financial crisis).

Spending Policy

The Investments Office and the Investment Committee worked together to recommend how much the endowment contributed each year to the University's budget. In 1992, the Investments Office

recommended that the Yale Corporation increase the endowment spending rate from 4.5% to 4.75%. In 1995, the Corporation approved a 5.0% "draw" and upped it to 5.25% in 2004.[13] The University thus benefited in two ways from the strength of its investment program: from the handsome returns and from a larger payout rate.

The Management of Marketable Securities

The Investments Office's philosophy—a focus on equity, diversification, inefficient markets, and outside managers with strongly aligned incentives—guided Yale's decisions in all of its actively managed asset classes. Swensen viewed the endowment's current target allocation of 7.5% in bonds and cash primarily as a liquidity reserve. The bond portfolio posted 1.2% 10-year annualized returns. Yale held U.S. government issues almost exclusively: Swensen doubted whether returns from U.S. corporate bonds adequately compensated investors for the added default risk and the callability of corporate issues. He was quite skeptical of foreign fixed-income securities as well.

Unlike the rest of its portfolio, the Investments Office managed its bond portfolio internally. Swensen believed that the government bond market was so efficient, and the spread between the performance of government bond fund managers so small, that it did not make sense to hire an outside manager. (In many cases, managers sought to boost returns by buying riskier bonds, defeating the purpose of holding bonds in the first place.) No attempt was made to add value through trading on interest rate movements. The endowment staff generated incremental returns only through modest bets on securities that were backed by the full faith and credit of the United States.

Events in the spring of 2020 significantly altered the bond portfolio's rebalancing process. As a result of severe disruption in the Treasury markets, bid-ask spreads widened dramatically. Yale realized that the monthly rebalancing process was uneconomic, as transaction costs were almost equivalent to the portfolio's returns in the very low-rate environment. Yale's internal team replaced bonds that matured with the necessary short-duration bonds of up to 18 months to preserve its distribution of maturities. Carrie Abildgaard, Director of Fixed Income, anticipated the suspension of full rebalancing to continue as long as markets remained disrupted.

Yale invested in U.S. common stocks, but it was a small target allocation of 2.25% of assets. The portfolio returned 12.8% annualized over 10 years. The domestic equities market was very efficient, and it was challenging for managers to generate reliable outperformance. Swensen and the team were convinced, however, that disciplined fundamental-based approaches, when intelligently applied, could generate reliable and superior long-run performance. Yale opted to invest with a small number of active equity managers who emphasized focused and differentiated strategies. Not surprisingly, none of Yale's managers used market timing or quantitatively-driven investment approaches. Xinchen Wang, Director of Domestic Equity, added, "Fee structures are important considerations as well. We try to incorporate market-based hurdles so that managers are paid performance fees only when they generate outperformance." Domestic equity managers tended to be smaller independent organizations that were owned by their investment professionals. Swensen worried that money managers at larger organizations tended to emphasize asset growth at the expense of performance and that ownership by a large institution reduced organizational stability and dampened incentives to perform.

A more diffuse category of publicly traded investments was an asset class called absolute-return strategies, which Yale "invented" in 1990 and to which the endowment had allocated 23.5% of its assets. These holdings included a variety of funds that exploited market inefficiencies. Swensen maintained the importance of "understanding the strategy of the hedge funds and investing in groups where there

is a clear return generation strategy and investors are treated fairly." Between 2010 and 2020, the absolute return portfolio posted a 5.3% annualized return.

Yale divided absolute-return funds into two broad categories: event-driven and value-driven. Event-driven strategies created hedged positions in mispriced securities and depended on a specific corporate event, such as a merger or bankruptcy settlement, to achieve targeted returns. Value-driven strategies relied on changing company fundamentals or increasing market awareness to drive prices toward fair value. Both strategies were expected to produce equity-like returns that were not highly correlated with any particular financial market. Importantly, the absolute return portfolio exhibited low beta to broad market indices. After the financial crisis, when absolute return produced a negative 9.3% return, Yale became more rigorous about estimating managers' betas and sought to target a specific range for its portfolio of 0.0 to 0.15 beta. If the portfolio's beta rose to an undesirable level, Yale would prioritize increasing its exposure to active short sellers or to market hedges.

Foreign equities, both developed and emerging, had a target allocation of 11.75% of endowment assets and offered a valuable source of diversification, as their returns tended to be only partially correlated with those of the U.S. equity market. Many of Yale's managers were regional specialists, although a few had global mandates. As Lisa Howie, Director of Foreign Equity, said, "We really look for places that have knowledgeable managers on the ground. We believe that focused, local investors can conduct deeper research, build stronger networks, and develop keener insights than those who cover multiple markets and spend limited time in the geographies where they invest."

Previously, Yale had difficulty in finding talented, research-driven emerging market managers with the proper incentives. Howie said, "Now there are many more stand-alone investment management firms. In part, this is due to a few very successful firms, which spawned the next generation of firms. China, India, and Brazil all have very vibrant communities of investment management firms." In 2005, Yale seeded Hillhouse Capital's founder and Yale alumnus, Lei Zhang, with $20 million in capital. After working in the Investments Office, translating Swensen's books into Chinese, and later working at a fund in which Yale was an investor, the young manager planned to return to China to work at an affiliate of a U.S.-based hedge fund. Instead, Yale became a cornerstone investor in his fund. The fund had done well for Yale, and Zhang became one of the most successful investment managers in China.

Yale's foreign equities portfolio had generated an annualized 15.8% return over the previous decade. Both the developed and emerging market subclasses outperformed. The developed market portfolio produced 13.6% 10-year annualized returns. Even more impressively, the emerging market portfolio produced 10-year annualized gains of 18.0%. Despite the growing macro-level concerns in several markets, Yale believed that emerging markets would continue to be less efficient and provide more opportunities for excess returns.

In 2020, new complications arose with the travel restrictions and social distancing necessitated by the COVID-19 pandemic. Yale's due diligence process emphasized spending in-person time with managers, which was essentially impossible during the pandemic, particularly for the foreign equity portfolio. Although sourcing did not stop, the pandemic lengthened the evaluation period for new investment managers.

The Management of Leveraged Buyouts

Yale had been among the first universities to invest in leveraged buyouts, entering into its first buyout partnership in 1973 (see **Exhibit 10**). The current allocation target was 17.5% for leveraged buyouts. Yale's LBO investment strategy was consistent with its overall investment philosophy. First,

the Investments Office placed a premium on building long-term relationships with a limited number of premier organizations. Yale's prestige, name, and long experience in investing made it a very desirable client and allowed it to invest in some well-regarded funds that might otherwise have been closed.

Yale emphasized organizations that took a value-added approach to investing. It shied away from funds that generated the bulk of their returns from simply buying assets at attractive prices, refinancing them, and flipping them. The philosophy was explicated in a discussion of buyout organizations: "While financial skill is a vital component of LBO investing, we seek firms that build fundamentally better businesses. Financial engineering skill is a commodity, readily available and cheaply priced. Value-added operational experience, however, is rare."[14] Yale sought "strong, cohesive, and hungry teams with proven ability to create value independent of the public market."[15] As a general rule, Yale gave considerable latitude to its firms to define the deals that they wanted to do.

Another key principle was to select organizations in which the incentives were properly aligned. For instance, Yale refused to invest in organizations affiliated with larger financial institutions. Such situations, the Investments Office believed, were fertile breeding grounds for conflicts of interest or insufficient incentives for the people actually doing the deals or both.

LBOs have been subject to a boom-and-bust cycle since at least the 1960s, with high returns attracting new investors who flooded money into the sector until returns deteriorated, whereupon they withdrew. But the unprecedented growth of the LBO industry appeared to have changed it in some permanent ways.

First was the scale on which buyout groups operated, where multibillion-dollar funds had become the norm (see **Exhibit 5** again for inflows into buyouts). As Tim Sullivan, Director of Leveraged Buyouts, noted, "Many LBO firms appear to have explicitly lowered their return hurdles in order to compete for transactions, particularly at the larger end of the market, pricing deals to yield returns in the mid-teens."[16] Even when investing in middle-market buyout groups, Yale often found itself progressively uncomfortable with the size of the funds being raised. Sullivan said, "Successful firms can raise more money which enables buying larger companies. Firms believe larger companies are better to own because they are in fact better, more professionalized companies. But, they have to pay more for the privilege of ownership, and better managed companies have less room for operational improvements. In addition, some firm's skillsets have been honed for smaller businesses and may not easily translate to bigger businesses. Consequently, returns may suffer as funds grow."

More generally, Yale noted with dismay that a number of groups were positioning themselves as "asset managers"—for instance, raising absolute-return, venture capital, mezzanine, energy, and real estate funds in addition to their core buyout funds. Related to this was concern about the substantial amount of undeployed capital in funds. Not only might this inflate deal valuations and reduce returns, but it might also allow underperforming firms to eke out an existence, further congesting the market.

The second major change involved the new classes of investors active in the industry. While overall fund-raising steeply declined during the financial crisis, investor appetite for LBOs had steadily increased since then, accompanied by growing demand amongst institutional investors for co-investment opportunities. Yale particularly disliked the tiering of the buyout industry, with large players demanding fee breaks and special deals.

In spite of the complications, Swensen and Sullivan believed that Yale should stay committed to the LBO portfolio for three reasons:

- Great returns. Yale's LBO portfolio had delivered an annual rate of return of 14.6% over the past decade. Since inception in 1973, the portfolio had returned 16.8%.
- Deep institutional knowledge. Swensen and Sullivan had worked together on the portfolio for more than three decades. The Investments Office noted that many of the funds managed by top-tier LBO organizations—for instance, Berkshire Partners, GTCR, and Roark—had continued to generate superior returns.
- Consistency. If Yale were to decide not to invest with a top-tier firm merely because the market was overheated, it might not be able to persuade the organization to include it when later market conditions were more favorable. As Sullivan concluded, if Yale were to alter its steady commitment to private equity and seek to time the market, top-tier firms "would not want Yale's unreliable money."[17]

An area of continuing interest was international LBOs. While many other institutional investors saw international buyouts as particularly promising, Yale eschewed the typical strategy of investing in large funds devoted to buyouts in Europe and Asia. In many cases, the office was unsure of the weight to put on the positive and negative feedback it gathered in its due-diligence process. Furthermore, many of the LBO firms in Europe and Asia were associated with larger financial institutions, which Yale wanted to avoid.

These challenges were even greater in emerging markets. The emerging *public* equity markets provided attractive opportunities along with a reasonable degree of liquidity, creating a high bar for emerging markets LBO investments. Howie said, "We find ourselves asking whether emerging market private equity funds are worth the fees, lack of transparency, and illiquidity when the public markets can be so compelling." In addition, the value-added private equity approach that Yale sought in its domestic private equity portfolio was more difficult to execute in emerging markets where foreign ownership constraints often precluded control investing, and it could be difficult to replace management or take other necessary action to improve troubled companies.

Furthermore, property rights, contract enforcement, corporate governance, disclosures, and minority shareholder rights—requisites for private deals—were still developing in many emerging markets. Frauds were not uncommon and could be difficult to detect, particularly for foreign investors. China presented additional obstacles in the form of capital controls, uncertain tax policies, fragmented private equity regulations across national and municipal authorities, and a lengthy list of industries in which foreign investment was prohibited or restricted. At the same time, however, private equity capital had flooded into emerging markets in recent years, driving up valuations. Could emerging market LBO funds provide attractive risk-adjusted returns?

Yale was deeply cautious when selecting LBO funds in emerging markets and sought the same qualities as it did in firms in developed countries: small entrepreneurial teams with operational experience on the ground, some co-investment, appropriate management and incentive fees, and a keen sense of where upside opportunities might lie.

The Management of Venture Capital

Yale's venture capital portfolio had grown considerably since it entered the asset class in 1976. The current target allocation to the asset class was 23.5%, and the portfolio posted 10-year annualized returns of 21.3%.

Previously, leveraged buyouts and venture capital were managed together as one private equity portfolio. In 2015, the Investments Office decided to carve out venture capital from the leveraged buyouts portfolio. This decision was made for several reasons. First, venture capital had surpassed expectations and grown to be a substantial allocation in the endowment. Second, venture capital and LBO funds had fundamentally different drivers of return. Third, the change allowed Yale to manage its venture capital portfolio better, both in the context of asset allocation decisions and in terms of internal resources.

Yale primarily chose to partner with prestigious Series A venture capital managers, such as Sutter Hill Ventures. The Investments Office found that its highest returns came from the groups in which the amount it could invest was severely constrained; put another way, the most exclusive clubs truly were the best! While Yale had made numerous investments into less well-known funds in hopes of backing the leaders of tomorrow, these had generated more mixed results. Although Yale would still consider new managers with a unique advantage, backing its winners had proved the best strategy.

Swensen mused that if he were starting an investment program today, he might avoid venture capital entirely. He reasoned that, for a newcomer, obtaining access to the best firms was nearly impossible and that, even if successful, the endowment was unlikely to receive an allocation that would be material to a $31.2 billion portfolio. Of course, Yale's long-standing relationships in the venture world put the university in a position different from that of a new entrant.

Despite impressive performance, the category was not without concerns. Yale remained skeptical of growth equity over the long term and was apprehensive of its early-stage managers raising larger funds to pursue the strategy. Matthew Mendelsohn, Director of Venture Capital, noted, "Many of our partners want to move up-market, but we think the risk-return profile is different in that segment of the market. We want to be supportive of our partners, so it is challenging to strike the right balance."

Yale paid close attention to the alignment of incentives to ensure that firms were not raising bigger funds for the purpose of "fee-gathering." Yale preferred an overall structure such that the firm could just cover its ongoing costs from the annual fee, earning essentially all of its economic returns from the "carry" tied directly to investment performance. This policy could at times be problematic: for instance, several of the most successful venture funds had dramatically increased their annual management fee income during the 1990s. While Yale would have liked to insist that the bulk of the compensation be linked to investment performance, in many cases it had been unable to persuade the partners to change. Some of these venture organizations were sufficiently attractive that the Investments Office participated in their funds anyway. In other cases, due to fundamental changes in the venture firm's investment strategy or organizational structure, Yale declined.[18]

Yale developed successful partnerships with venture capital funds in emerging markets, mostly in China. However, evaluating and selecting these venture capital managers was extremely challenging, perhaps more severe than in almost any other asset class. Most desirable were groups that had strong U.S. ties and, ideally, ties to Yale—and these were not easy to find even in China, the emerging market Yale knew best. China was at the forefront of many innovative technologies, and Yale was optimistic about early-stage venture capital there. However, rising political tensions between the U.S. and China along with COVID-19 travel restrictions created complexities in the due diligence process.

The team agreed that foreign exposure was important in the venture capital portfolio, but were the numerous opportunities in China worth the risk? And, where were conditions right for the next burgeoning venture capital environment? "In general, we like funds and companies that control their own destiny, and in some places the macro environment is so tough you can't control your own

destiny. We prefer a place where a fund will do okay if the macro is tough and really well if it is favorable", said Howie.

The Management of Real Estate

Another important asset class was real estate, which had a target allocation of 9.5%. From 2010 to 2020, the real estate portfolio generated 9.7% annualized returns. The Investments Office believed that properly managed real estate provided an interesting set of investment opportunities. The returns from real estate tended to be uncorrelated with those from marketable common stock and, in the long run, should produce returns protected from inflation. Most important, though, real estate was an inefficient, cyclical market in which Yale could generate very attractive returns if it found the right managers. As in other asset classes, Yale concentrated on pure equity investments, avoiding mortgages and other debt. The Investments Office shunned managers who were financial advisers that bought existing buildings with stable rent rolls and applied financial engineering. Instead, the office sought to establish relationships with real estate operators who had a competitive advantage, either by property type or market, and preferably a focus on an out-of-favor sector.

Historically, Yale's real estate portfolio had consisted primarily of a single Manhattan office building at 717 Fifth Avenue, a direct investment that performed very well. Yale paid $14 million for a 50% interest in 1978 and $47 million for the remaining 50% in 1994. Yale's October 2002 sale of 717 Fifth Avenue at $611 per square foot represented one of the highest prices ever paid for an arm's-length sale of a Manhattan office property. Over the 24-year holding period, Yale realized a 19.5% per annum return on its investment. In spite of the strong results, the challenges in managing 717 Fifth Avenue reinforced Yale's strong preference for external management of endowment assets.

Beginning around 1990, Yale came to believe that real estate presented a compelling opportunity and accordingly began increasing its real estate investments. Yale chose to focus on deliberately contrarian segments of the real estate market that most other investors avoided. The Investments Office sought out partners who targeted distressed sellers and possessed the operating expertise to implement value-added strategies that realized substantial returns over the medium term.

Yale encountered some challenges in implementing its real estate strategy. The Investments Office believed that the institutional real estate industry had significant misalignment or conflicts of interest and that most firms were compensated through transaction fees or fees based on assets under management, rather than by shared profits. These firms thus had every incentive to lock up their investors' capital for long periods and focus on asset accumulation and retention rather than generation of superior returns. Instead, Yale wanted the principals to make a significant cash co-investment in the fund; it wanted an appropriately sized fund; and it wanted most of the principals' compensation to come at the end of the fund and to be linked to investors' returns.

Over time, working their networks, the Investments Office staff managed to find a number of independent firms with excellent real estate operating skills that were eager to forge such a relationship. Unlike in private equity, where Yale invested in the premier institutional funds, its real estate funds were not well known, even by knowledgeable real estate investors. Yale was often the lead investor in these funds, with a sizable percentage of the limited partnership interest.

After the 2008 financial crisis, it became readily apparent that Yale's managers had exercised a wide range of discipline heading into the downturn. The core managers survived intact, while managers engaged in development and highly leveraged strategies suffered. As a consequence, Yale reviewed its real estate managers and pared its portfolio to focus on its highest conviction relationships.

The COVID-19 pandemic created new challenges for the real estate portfolio. The pandemic had drastically different implications by industry, and real estate segments did not move in unison. Real Estate Director Alan Forman noted, "Social distancing and work-from-home policies have created uncertainty about the future of densely-populated city centers and the need for large, downtown office buildings. The travel and hospitality industries were hit particularly hard and many hotels suffered. In contrast, the second home market, especially in less populated areas, has done well during a downturn when it would typically underperform. Unexpected outcomes during the pandemic make me think it will be difficult to predict how behaviors might change once again after it is over." Despite the uncertainty, Yale felt better positioned due to its efforts after 2008 to prune real estate managers and reduce risks in the portfolio.

The Management of Natural Resources

Yale's natural resources portfolio contained the University's oil-and-gas, timberland, and agriculture partnerships. The sensitivity of natural resources to commodity prices provided an important hedge against inflation and enhanced the Investments Office's ability to cushion the impact of rising prices on the university. To date, Yale had focused most of its efforts on disciplined purchases of cash-generative investments, such as energy reserve acquisition programs and managed timberland. Recently, Yale had started to build exposure to agricultural assets.

It was difficult to find well-designed oil-and-gas partnerships led by attractive managers. Many of the partnerships appeared to be in the hands of agents, who were compensated primarily on the basis of arranging deals. A number of operators seemed to get rich even if their clients did not. Furthermore, assessing the skills of the general partners in these funds was often difficult. In many cases, individuals raised funds on the basis of their participation in earlier successful partnerships, but determining which partner had been responsible for a key discovery or production success was challenging.[19] Yale's general impression was that investment opportunities and partnerships with sterling track records, unblemished reputations, and proper deal structures were quite uncommon.

As a result, Yale's investments in oil and gas tended to emphasize two different models. The first focused on partnerships in the business of acquiring existing oil and gas fields and enhancing their operations (known as reserve acquisitions). In contrast to the risky world of exploration in which positive outcomes were hard to predict and even harder to repeat, producing assets were straightforward to underwrite and provided stable income that was sensitive to inflation. The other approach applied a private equity investment model, whereby Yale invested in partnerships pursuing equity investments in oil-and-gas and energy service companies.

In recent years, Yale opted to reduce its investment in traditional private equity funds and instead backed oil-and-gas management teams directly through co-investment or primary equity investments. Management teams were nimbler and more opportunistic than private equity funds and could create value through patient capital allocation (investing with a long time horizon), superior operations, and the prudent development of proved reserves.

Yale invested in a number of forestland partnerships focused on sustainable harvesting of softwood and hardwood forestland in the United States. Although the initial investments had been made at very attractive valuations, new investors had flooded the market, resulting in very few opportunities to invest in timberland at compelling valuations. Yale's allocation to timberland declined in the past decade, as the University's managers were more inclined to be sellers than buyers. In 2020, Yale looked to rebuild their exposure in the category.

The natural resources portfolio briefly contained metals and mining partnerships. These investments halted as returns did not compensate for the high risks in the sector. Instead, Yale turned to investing in agriculture. Farmland was appealing due to the optionality of the land, which could be used for either traditional farming purposes or new practices, such as building solar fields. The allocation to agriculture was initially small but was quickly growing.

The endowment had a current target allocation of 4.5% for natural resources, which was less than the average endowment. Even though longer term performance was remarkable, over the past 10 years, Yale's natural resources portfolio had generated a somewhat disappointing annual return of 4.4%. In recent years, returns in oil-and-gas and forestland had suffered sharply, but the portfolio still managed to outperform relevant benchmarks.

Moreover, natural resource investments were not without controversy: like its peers, Yale's investments in this area had been criticized by campus activists. Before departing, Takahashi commented, "Many natural resources are closely linked to sensitive issues, including food, water, the environment, and safe working conditions. If we increase our exposure to these assets, we will have to proceed with our usual care to ensure that we back the right managers."

Liquidity

The "100-year storm" seemed to occur nearly every decade with "Black Monday" in 1987, the dot-com bubble bursting in 2000, the 2008 financial crisis, and the 2020 COVID-19 pandemic. These events underscored the need to be prepared. "We do not want be forced to sell assets at distressed prices to meet university spending needs or fund capital calls. We want to be able to access capital to play offense in these situations", said Swensen.

During the dot-com bubble in the late 1990s, responding to the need to generate liquidity to support hedging activities, the endowment developed a number of sources of liquidity that did not require sales of investment positions. Previously, Yale used bonds as collateral for short-term loan agreements. These agreements had been a major advantage for Yale in previous times of crisis-imposed liquidity constraints. However, the availability of repurchase agreements had become unreliable recent years, particularly over quarter ends, so Yale had paused the use of these agreements.

Yale had access to commercial paper facilities that had been established in the early 1990s by the Investments Office. Set up to facilitate operating transactions, capital projects, and endowment liquidity, the commercial paper facility, which was backed by fully committed bank lines of credit, provided Yale with access to $2 billion of funds during the 2008 financial crisis. In addition, the university retained the ability to tap long-term financing markets for funds, and did so in both the financial crisis and the pandemic meltdown. Decisions around the issuance of long-term debt were made to ensure the university had sufficient operational capital over an intermediate time horizon without restricting the activities of the endowment.

Lastly, Yale had the ability to sell investment positions in its in-house bond portfolio. Yale viewed cash generation as one of the primary purposes of the bond portfolio, and this was the preferred way to satisfy immediate liquidity needs. The endowment resisted selling partnership stakes in illiquid investments, as these could be subject to substantial discounts, especially in times of crisis.

After the 2008 crisis, the Investments Office had adopted a number of new policies. Yale extended a model, initially developed by Dean Takahashi and Seth Alexander,[20] to forecast the cash flow patterns of every leveraged buyout, venture capital, real estate, and natural resource fund in its portfolio. This

helped the Investments Office anticipate the impact of its illiquid commitments on cash flow requirements over the coming years. As a result, the Investments Office decided to limit the share of illiquid assets to 50% of the portfolio. Swensen said, "This might seem high, but at our size we have more options for liquidity than do smaller funds." At the 50% illiquid assets ceiling, the Investments Office estimated that the endowment would produce slightly lower rates of return, but also lower volatility than would otherwise be the case. There would be little change to the portfolio's ability to withstand catastrophic events.

Swensen commented, "Liquidity doesn't show up in a mean-variance model. You have to think about it very carefully." He still believed in illiquidity as the inevitable cousin of diversification and high-return investment opportunities. "If you sell all your risky stuff just because it's risky, you'll end up holding lots of Treasuries yielding next to nothing," he said.

Yale as a Limited Partner

Yale was deeply committed to its strategy of partnering with best-in-class outside managers. One of these managers, Sutter Hill Ventures, was the oldest and one of the most renowned venture capital firms in Silicon Valley. In 1991, G. Leonard Baker (Yale '66), Partner at Sutter Hill Ventures, met with Swensen and Takahashi. At that time, Sutter Hill Ventures only had one limited partner and wanted to bring in new sources of capital. Shortly after the meeting, Yale invested, and Baker joined the Yale Investment Committee. This partnership paved the way for other prestigious endowments, including M.I.T., Princeton, and Stanford, to invest.

Baker commented, "Yale has always been a great LP throughout our long partnership." A stable LP base has been advantageous for Sutter Hill Ventures as it has allowed them to invest over long time horizons in an evergreen fund and to be contrarians. As Baker said, "The more people you have to explain yourself to, the more likely you are going to be exposed to conventional wisdom." This strategy had led Sutter Hill to massive successes, most recently in backing Snowflake: at the time of Snowflake's 2020 IPO, Sutter Hill's gains were projected to be the largest from any investment by a venture firm.[21]

Baker also reflected on the trend of venture capital funds growing larger and moving up-market. He noted, "There are two types of funds: asset gatherers and investors. Asset gatherers want to earn higher fees based on increasing the assets under management, and investors want to achieve superior returns on capital to share with their LPs. Yale is particularly acute at distinguishing between these two categories." Yale's experience in sourcing high quality managers and its deep relationships were a differentiated advantage over LPs new to emulating the Yale model. As Amy Chivetta, Director in Leveraged Buyouts and Venture Capital, said, "Thanks to our decades-long history of investing in external managers, we have encountered many of the issues that investment firms face as they develop. We want to be the first call for our managers when they need guidance."

Future Challenges

The market still posed fundamental challenges. Buyout funds, with their massive capital overhangs, were bidding up valuations; venture funds were pursuing companies with multibillion-dollar valuations instead of innovative tech start-ups. Emerging markets required time-intensive, careful evaluation, which was especially challenging under COVID-19 travel restrictions. Work-from-home policies and social distancing shifted real estate markets in city centers, and it was unclear how long these behaviors would endure.

Looking beyond the short run, Swensen wondered about the coming challenges that the Yale endowment would face. Yale's continuing expansion had increased the demands placed on the endowment—and the illiquid asset allocation ceiling reduced some of the endowment's most potent tools for creating returns. Were they at the right balance? Did the projected approach to the new target make sense? How should they think about the difficulty in determining precise valuations for the endowment?[22] In the long run, how should they think about the issues of risk and illiquidity? And could a strategy that worked so well for Yale when it was a billion-dollar fund continue to succeed as the pool of assets grew?

Large endowments have always faced public scrutiny, but pressure had escalated in recent years. Swensen joined the Yale endowment during the movement to divest from South Africa in the 1980s. Since then, there had been several other national divestiture movements including tobacco, fossil fuels, and private prisons. Congressional leaders had pressured endowments to boost diversity in the asset management industry, a worthy but often challenging goal. In addition, the 2017 so-called "endowment tax," a 1.4% excise tax on net investment income at private universities, and increased regulations, particularly aimed at private equity, could create new challenges for endowments. How would policy changes alter endowment operations? And how should Yale and other endowments respond to external pressures on investment strategy?

Finally, how should Yale think about Swensen's seemingly unique abilities? Several other endowments had done well with adopting and tweaking the Yale model, but, as Baker said, "There is a difference between creating the principles and following the recipe." Swensen not only drove performance; his leadership and values were central to the office culture. As John Ricotta, Director of Natural Resources, said, "Working with David is one of the best parts of the job. I am always learning from him and being around greatness makes you want to be great." This sentiment was echoed by all Directors at the office. How much of the success at Yale was tied to Swensen? With more endowments adopting its model, how could Yale keep its competitive advantage?

Exhibit 1 Asset Allocations of Yale Endowment, 1985-2020

	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003
Domestic Equity	61.6%	63.5%	61.7%	56.8%	53.2%	48.0%	30.7%	27.5%	23.9%	21.2%	21.8%	22.8%	21.5%	19.2%	15.1%	14.2%	15.5%	15.4%	14.9%
Foreign Equity	6.3%	8.6%	10.8%	14.0%	15.4%	15.2%	14.8%	15.3%	16.5%	14.6%	12.5%	12.5%	12.8%	12.1%	11.1%	9.0%	10.6%	12.8%	14.6%
Cash & Bonds	20.4%	17.7%	16.7%	17.1%	16.6%	22.1%	22.4%	23.2%	22.6%	17.1%	14.0%	13.9%	12.3%	7.6%	11.1%	17.5%	16.0%	10.3%	9.5%
Real Assets	8.5%	7.5%	7.2%	7.7%	8.7%	8.0%	7.9%	7.1%	6.0%	8.9%	13.5%	11.5%	11.5%	13.0%	17.9%	14.9%	16.8%	20.5%	20.9%
Private Equity	3.2%	2.7%	3.6%	4.4%	6.1%	6.7%	8.3%	10.4%	14.4%	18.1%	17.2%	18.5%	18.6%	21.0%	23.0%	25.0%	18.2%	14.4%	14.9%
Absolute Return	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	15.9%	16.5%	16.6%	20.1%	21.0%	20.7%	23.3%	27.1%	21.8%	19.5%	22.9%	26.5%	25.1%

	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	Current (2021) Target Allocation
Domestic Equity	14.8%	14.1%	11.6%	11.0%	10.1%	7.5%	7.0%	6.7%	5.8%	5.9%	3.9%	3.9%	4.0%	3.9%	3.5%	2.7%	2.3%	2.3%
Foreign Equity	14.8%	13.7%	14.6%	14.1%	15.2%	9.8%	9.9%	9.0%	7.8%	9.8%	11.5%	14.7%	14.9%	15.2%	15.3%	13.7%	11.4%	11.8%
Cash & Bonds	10.9%	6.8%	6.3%	5.9%	0.1%	2.1%	4.4%	2.8%	6.6%	6.5%	8.4%	7.7%	7.2%	5.8%	4.5%	8.4%	13.7%	7.5%
Real Assets	18.8%	25.0%																
Real Estate			16.8%	18.1%	18.9%	20.6%	18.7%	20.2%	21.7%	20.2%	17.6%	14.0%	13.0%	10.9%	10.3%	10.1%	8.6%	9.5%
Natural Resources			11.0%	9.0%	10.4%	11.5%	8.8%	8.7%	8.3%	7.9%	8.2%	6.7%	7.9%	7.8%	7.0%	4.9%	3.9%	4.5%
Private Equity	14.5%	14.8%	16.4%	18.7%	20.2%	24.3%												
LBO							23.4%	24.8%	24.3%	21.9%	19.3%	16.2%	14.7%	14.2%	14.1%	15.9%	15.8%	17.5%
Venture Capital							6.8%	10.3%	11.0%	10.0%	13.7%	16.3%	16.2%	17.1%	19.0%	21.1%	22.6%	23.5%
Absolute Return	26.1%	25.7%	23.3%	23.3%	25.1%	24.3%	21.0%	17.5%	14.5%	17.8%	17.4%	20.5%	22.1%	25.1%	26.1%	23.2%	21.6%	23.5%

Source: Yale University documents.

Note: Real Assets was split into Natural Resources and Real Estate to set the fiscal year 2012 targets. Private Equity was split into Venture Capital and Leveraged Buyouts (LBO) to set the fiscal year 2016 targets.

Exhibit 2 Asset Allocations of Large University Endowments, 1985-2019

	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002
Domestic Equity	51.5%	52.1%	53.8%	50.2%	46.1%	45.3%	43.5%	44.4%	43.0%	41.6%	42.8%	41.1%	42.3%	39.9%	36.3%	36.5%	30.1%	29.3%
Foreign Equity	2.0%	2.6%	3.0%	5.2%	6.6%	6.6%	7.8%	8.1%	10.2%	14.3%	15.2%	14.0%	15.9%	14.7%	14.5%	15.0%	13.9%	14.3%
Bonds	26.4%	28.3%	26.0%	26.2%	27.5%	29.2%	30.2%	30.7%	26.9%	22.4%	17.5%	20.1%	18.0%	15.9%	15.0%	17.4%	17.8%	23.1%
Cash	10.8%	8.8%	8.9%	7.7%	6.9%	6.9%	6.1%	5.2%	4.6%	3.2%	4.1%	3.2%	3.1%	2.5%	3.0%	2.8%	2.5%	-0.2%
Real Assets	4.8%	4.9%	5.2%	4.3%	5.1%	4.4%	4.2%	3.7%	3.7%	4.2%	5.1%	5.2%	5.2%	6.9%	6.6%	4.7%	6.3%	5.9%
Private Equity	2.7%	1.9%	2.0%	5.8%	6.6%	6.2%	6.2%	5.9%	6.6%	7.7%	8.0%	8.1%	7.2%	8.2%	11.1%	13.9%	12.1%	9.9%
Absolute Return	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.6%	0.8%	3.3%	5.2%	6.2%	6.7%	6.9%	10.1%	11.3%	8.3%	13.2%	14.7%
Other	1.8%	0.0%	1.1%	0.6%	1.2%	1.3%	1.4%	1.2%	1.7%	1.5%	1.1%	1.6%	1.5%	2.0%	2.1%	1.2%	4.1%	3.1%

	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Domestic Equity	28.6%	28.3%	24.5%	20.7%	20.6%	15.1%	14.0%	11.0%	12.0%	12.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	11.2%
Foreign Equity	14.2%	16.1%	17.4%	21.4%	22.9%	20.8%	12.0%	15.0%	16.0%	15.0%	17.0%	18.0%	19.0%	19.0%	19.0%	19.0%	20.1%
Bonds	19.4%	16.0%	16.2%	14.2%	11.9%	11.3%	10.0%	10.0%	9.0%	9.0%	8.0%	8.0%	7.0%	7.0%	7.0%	7.0%	7.9%
Cash	0.4%	2.6%	1.1%	0.4%	0.3%	-0.6%	3.0%	4.0%	3.0%	3.0%	3.0%	4.0%	4.0%	3.0%	4.0%	3.0%	4.4%
Real Assets	11.0%	9.7%	10.7%														
Real Estate				6.0%	6.2%	7.5%	7.9%	6.6%	7.8%	9.0%	8.0%	7.0%	7.0%	7.0%	6.0%	5.0%	6.5%
Natural Resources				5.5%	6.3%	8.3%	7.3%	8.4%	9.0%	9.0%	8.0%	8.0%	7.0%	8.0%	8.0%	7.0%	6.6%
Private Equity	8.7%	8.6%	9.3%	9.9%	11.3%	14.1%	18.3%										
LBO								15.0%	15.6%	16.0%	15.0%	12.0%	12.0%	12.0%	12.0%	11.0%	13.6%
Venture Capital								4.2%	4.8%	5.0%	5.0%	5.0%	7.0%	7.0%	7.0%	8.0%	9.0%
Absolute Return	17.3%	17.9%	19.9%	21.1%	20.2%	23.1%	27.5%	25.8%	22.8%	22.0%	22.0%	20.0%	23.0%	22.0%	22.0%	20.0%	20.5%
Other	0.4%	1.0%	0.8%	0.8%	0.6%	0.5%	0.0%	0.0%	0.0%	0.0%	1.0%	5.0%	1.0%	2.0%	2.0%	7.0%	0.0%

Source: Compiled from National Association of College and University Business Officers, NACUBO-TIAA Study of Endowments (Washington, National Association of College and University Business Officers, 2019) (and earlier years).

Note: Asset allocations are on June 30 of each year.

Large funds are defined as those with more than $1 billion in assets after 1998, as those with more than $400 million in assets in 1988 through 1997, and as those with more than $200 million in assets in 1985 through 1987.

Private equity includes venture capital, buyouts, and (through 2005) oil and gas.

Absolute return includes hedge funds, high-yield bonds, distressed securities, and event arbitrage.

Funds are weighted equally in calculating average allocations through 1987; funds are weighted by size in calculating average allocations in later years.

1985–1987 classifications may not be completely analogous to those in other years.

Exhibit 3 Asset Allocations of All University Endowments, 1985-2019

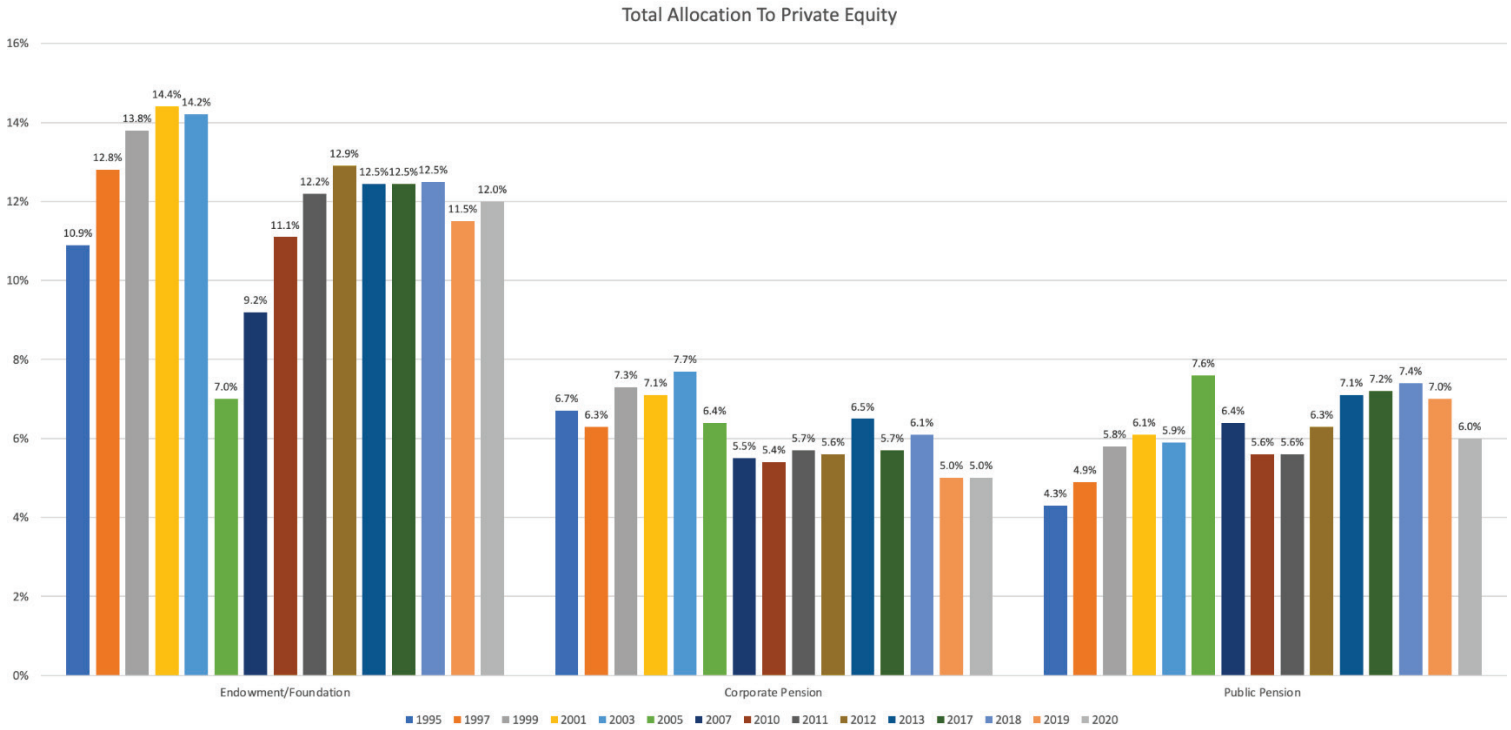
	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002
Domestic Equity	46.1%	48.7%	51.4%	46.4%	48.5%	48.1%	47.1%	47.1%	48.5%	47.2%	49.2%	51.6%	52.6%	52.4%	53.7%	41.4%	49.5%	47.3%
Foreign Equity	0.8%	1.1%	1.6%	1.5%	1.8%	2.4%	2.4%	3.2%	4.2%	7.5%	9.5%	9.5%	11.2%	11.0%	10.6%	14.1%	9.9%	10.1%
Bonds	30.6%	30.6%	30.8%	33.8%	32.2%	33.9%	35.3%	35.3%	34.4%	32.2%	28.3%	27.3%	25.2%	24.9%	23.1%	21.1%	24.9%	26.9%
Cash	14.5%	13.1%	12.6%	14.2%	13.0%	10.9%	10.1%	9.9%	7.6%	7.1%	6.5%	5.4%	4.8%	4.1%	4.0%	3.5%	4.1%	3.9%
Real Assets	4.2%	3.9%	2.2%	2.5%	2.7%	2.9%	2.9%	2.3%	2.1%	2.1%	2.3%	2.0%	2.0%	2.2%	2.1%	3.0%	2.8%	3.1%
Private Equity	0.7%	0.6%	0.7%	0.9%	1.1%	1.0%	1.1%	0.9%	1.1%	1.2%	1.3%	1.4%	1.4%	1.4%	2.2%	8.7%	2.4%	2.0%
Absolute Return	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.3%	1.1%	1.8%	2.0%	2.2%	2.4%	3.6%	3.8%	7.0%	4.2%	5.1%
Other	3.1%	2.0%	0.7%	0.6%	0.7%	0.7%	0.9%	1.0%	0.9%	0.9%	0.9%	0.5%	0.4%	0.4%	0.5%	1.1%	2.1%	1.6%

	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Domestic Equity	47.7%	48.9%	45.8%	42.4%	40.5%	34.9%	18.0%	15.0%	16.0%	15.0%	16.0%	17.0%	16.0%	16.0%	16.0%	16.0%	14.1%
Foreign Equity	9.5%	11.0%	12.7%	15.3%	17.0%	17.0%	14.0%	16.0%	17.0%	16.0%	18.0%	19.0%	19.0%	19.0%	20.0%	20.0%	21.1%
Bonds	25.9%	22.1%	21.5%	20.3%	18.6%	19.2%	13.0%	12.0%	10.0%	11.0%	10.0%	9.0%	9.0%	8.0%	8.0%	8.0%	9.5%
Cash	4.0%	3.7%	3.5%	3.4%	3.5%	3.9%	4.0%	5.0%	4.0%	4.0%	3.0%	4.0%	4.0%	4.0%	4.0%	3.0%	4.3%
Real Assets	3.2%	3.4%	4.0%														
Real Estate				3.5%	3.5%	4.1%	6.1%	5.2%	6.4%	7.0%	7.0%	6.0%	6.0%	6.0%	6.0%	5.0%	5.9%
Natural Resources				1.5%	1.6%	2.2%	6.1%	7.3%	8.0%	8.0%	7.0%	7.0%	7.0%	7.0%	7.0%	7.0%	6.0%
Private Equity	2.1%	2.1%	2.4%	2.8%	3.2%	4.4%	14.3%										
Leveraged Buyouts								12.5%	12.7%	13.0%	12.0%	11.0%	10.0%	11.0%	11.0%	10.0%	12.3%
Venture Capital								3.1%	3.7%	4.0%	4.0%	4.0%	5.0%	6.0%	6.0%	6.0%	7.6%
Absolute Return	6.1%	7.3%	8.7%	9.6%	10.6%	12.9%	24.5%	23.9%	22.2%	21.0%	22.0%	20.0%	22.0%	21.0%	20.0%	19.0%	19.1%
Other	1.6%	1.6%	1.4%	1.4%	1.4%	1.5%	0.0%	0.0%	0.0%	1.0%	1.0%	3.0%	2.0%	2.0%	2.0%	6.0%	0.0%

Source: Compiled from National Association of College and University Business Officers, NACUBO-TIAA Study of Endowments (Washington, National Association of College and University Business Officers, 2019) (and earlier years).

Notes: Asset allocations are on June 30 of each year.
Private equity includes venture capital, buyouts, and (through 2005) oil and gas.
Absolute return includes hedge funds, high-yield bonds, distressed securities, and event arbitrage.
Funds are weighted equally in calculating average allocations through 2008; funds are weighted by size in calculating average allocations in later years.
1985 and 1986 classifications may not be completely analogous to those in other years.

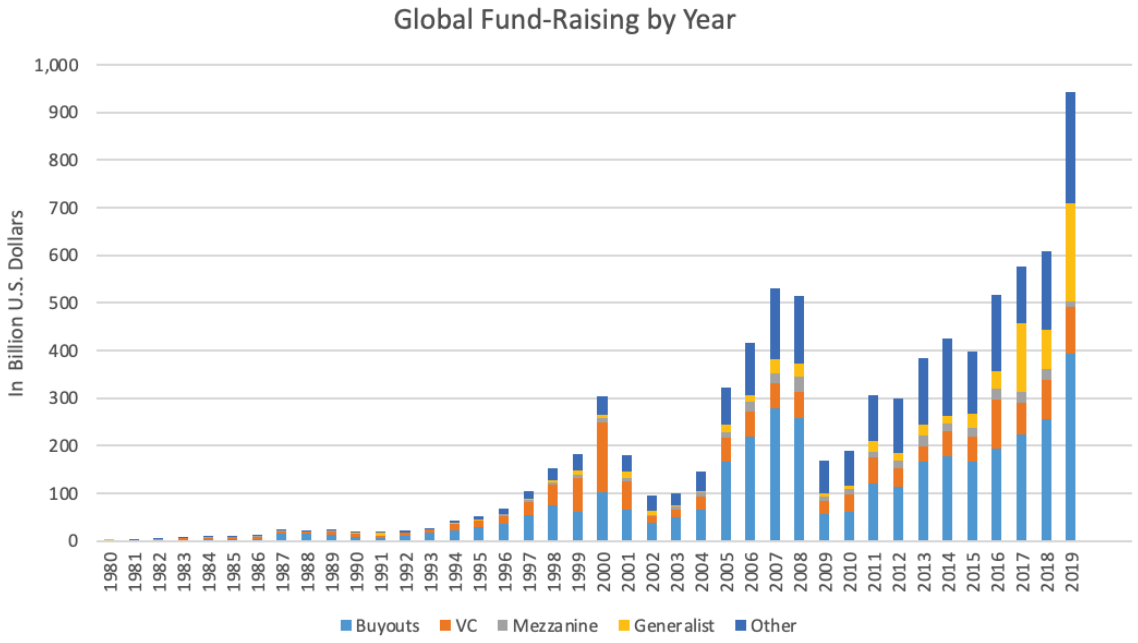
Exhibit 4 Asset Allocations of Major Pension Funds and Endowments, 1995-2020



Source: Data from 2020 come from Preqin, *2020 Preqin Global Private Equity and Venture Capital Report,* (London; Private Equity Intelligence Ltd. 2020). Data from 2019 come from Preqin, *2019 Preqin Global Private Equity and Venture Capital Report,* (London; Private Equity Intelligence Ltd. 2019). Data from 2018 and 2013 come from Preqin, *2018 Preqin Global Private Equity and Venture Capital Report,* (London; Private Equity Intelligence Ltd. 2018). Data from 2017 come from *2017 Preqin Global Private Equity and Venture Capital Report,* (London; Private Equity Intelligence Ltd. 2017). Data for 2010 and 2012 come from Preqin, *2014 Preqin Global Private Equity Review,* (London; Private Equity Intelligence Ltd. 2014). The balance was compiled from Russell Investment Group, *2007–2008 Russell Investments' Survey on Alternative Investing*, 2007; Russell Investment Group, *2005–2006 Russell Survey on Alternative Investing*, 2005; and Goldman, Sachs & Co. and Frank Russell Capital Inc., *2003 Report on Alternative Investments by Tax-Exempt Organizations,* 2003 (and earlier years).

Notes: The data for 2001 through 2007 are for North American institutions only; other years are for global institutions. Data from 2019 and 2020 are median target allocations; other years are averages. Endowment and foundations data for 2013-2020 was reported separately and the categories were averaged.

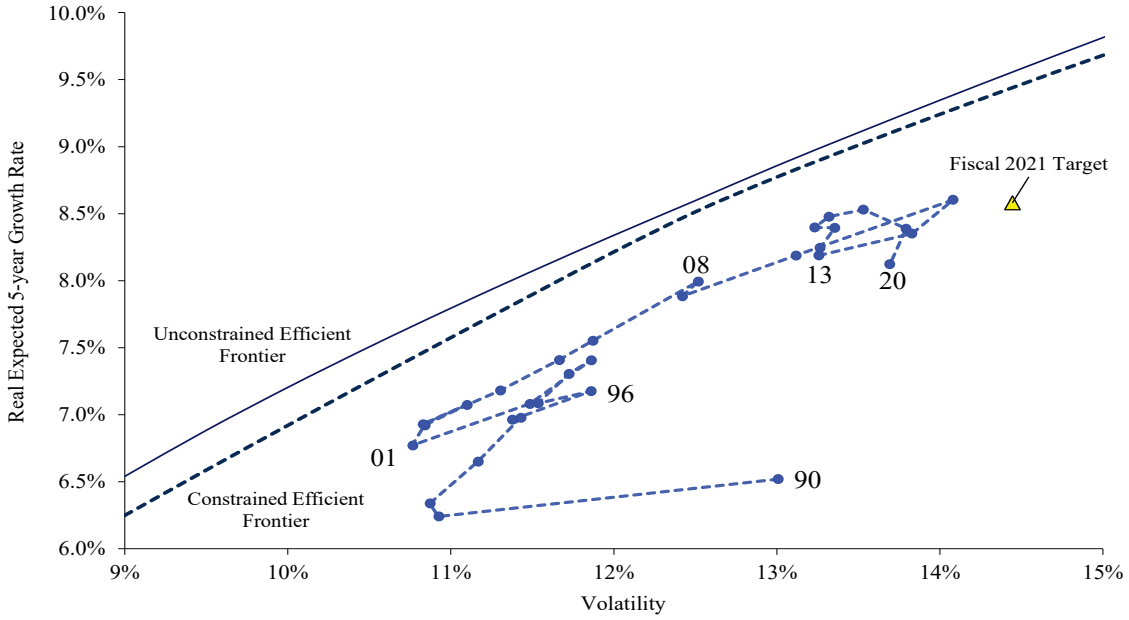
Exhibit 5 Global Private Equity Fund-Raising, by Fund Type, 1980-2019



Source: Casewriters, based on data from Thomson Reuters, accessed December 3, 2020.

Note: Generalist funds invest in both venture capital and buyouts at all stages. Venture Capital includes seed stage, early stage, later stage and balanced stage funds. Other includes energy funds, distressed debt funds, fund-of-funds, secondary funds, special situations funds, and real estate funds (core, opportunistic, and value-add).

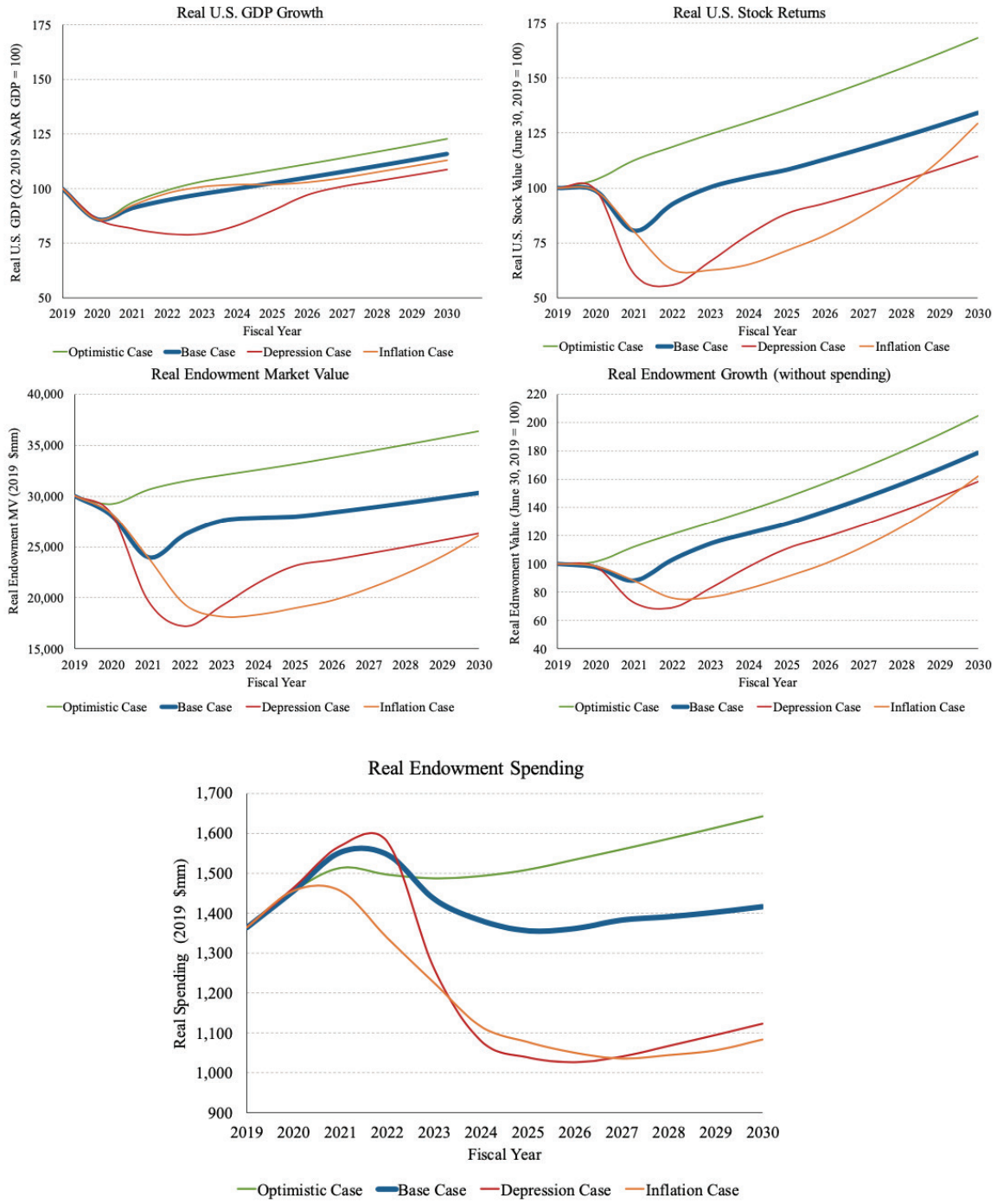
Exhibit 6 Yale's Historical Risk-and-Return Profile: Yale Moves toward Risk-Return Efficiency



Source: Yale University documents.

Note: Growth rate and volatility metrics since 2018 are presented net of the Endowment's excise tax.

Exhibit 7 COVID-19 Scenario Analysis



Source: Yale University documents.

Exhibit 8 Returns of All University Endowments, Yale Endowment, and Benchmark Indexes, Fiscal Years 1980-2020 (%)

	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994
Equal-weighted mean	12.9%	14.9%	-0.4%	40.5%	-2.4%	25.4%	26.6%	13.7%	1.3%	14.1%	9.6%	7.2%	13.1%	13.3%	2.9%
Cambridge Mean	13.6%	16.2%	-2.4%	44.7%	-2.8%	26.5%	28.8%	15.2%	1.2%	14.1%	9.9%	6.9%	14.1%	14.2%	3.8%
Yale	18.7%	22.7%	-4.5%	50.1%	-0.2%	25.8%	36.0%	22.8%	-0.2%	17.3%	13.1%	2.0%	13.4%	17.2%	11.9%
S&P 500	17.0%	20.4%	-11.5%	60.9%	-4.8%	30.7%	35.6%	25.1%	-7.0%	20.5%	16.5%	7.4%	13.4%	13.6%	1.4%
Wilshire 5000	19.2%	25.2%	-15.0%	66.5%	-8.7%	31.2%	35.2%	20.1%	-5.9%	19.5%	12.8%	7.0%	13.9%	16.1%	1.2%
Government bond index	3.8%	-4.1%	13.4%	29.1%	1.8%	28.7%	20.7%	4.7%	7.5%	12.3%	7.1%	10.2%	14.2%	13.1%	-1.5%
Consumer price index	14.3%	9.7%	7.2%	2.5%	4.3%	3.7%	1.8%	3.7%	4.0%	5.2%	4.7%	4.7%	3.0%	3.0%	2.5%

	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
Equal-weighted mean	15.5%	17.2%	20.4%	18.0%	10.8%	12.1%	-3.5%	-6.2%	3.2%	15.3%	9.3%	10.8%	17.2%	-3.0%	-18.7%
Cambridge Mean	15.7%	17.8%	20.5%	17.9%	12.0%	18.8%	-3.3%	-5.5%	3.2%	16.8%	12.1%	13.5%	20.0%	-1.2%	-19.9%
Yale	15.7%	25.6%	21.8%	18.1%	12.2%	41.0%	9.2%	0.7%	8.8%	19.4%	22.3%	22.9%	28.0%	4.5%	-24.6%
S&P 500	26.1%	26.0%	34.7%	30.2%	22.8%	7.2%	-14.8%	-18.0%	0.3%	19.1%	6.3%	8.6%	20.6%	-13.1%	-26.2%
Wilshire 5000	24.7%	26.2%	29.3%	28.9%	19.6%	9.5%	-15.3%	-16.6%	1.3%	21.3%	8.2%	9.9%	20.5%	-12.5%	-26.4%
Government bond index	12.8%	4.7%	7.8%	11.3%	2.7%	4.3%	11.1%	8.2%	13.1%	-0.7%	7.3%	-1.5%	6.0%	7.2%	5.3%
Consumer price index	3.0%	2.8%	2.2%	1.6%	2.0%	3.7%	3.2%	1.1%	1.9%	3.2%	2.5%	4.2%	2.7%	4.9%	-1.2%

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	Annualized
Equal-weighted mean	11.9%	19.2%	-0.3%	11.7%	15.5%	2.4%	-1.9%	12.2%	8.2%	5.3%	1.8%	9.2%
Cambridge Mean	11.9%	20.1%	-0.1%	11.9%	16.7%	3.0%	-2.5%	13.1%	9.3%	5.5%	2.2%	10.0%
Yale	8.9%	21.9%	4.7%	12.5%	20.2%	11.5%	3.4%	11.3%	12.3%	5.7%	6.8%	13.7%
S&P 500	14.4%	30.7%	5.4%	20.6%	24.6%	7.4%	4.0%	17.9%	14.4%	10.4%	7.5%	11.6%
Wilshire 5000	15.7%	32.0%	4.0%	21.1%	24.9%	7.1%	3.0%	18.5%	14.7%	9.1%	6.8%	11.5%
Government bond index	9.7%	3.7%	8.8%	-0.6%	4.3%	1.7%	6.7%	-0.4%	-0.6%	8.5%	10.0%	7.4%
Consumer price index	1.2%	3.5%	1.7%	1.8%	2.1%	1.0%	1.0%	1.6%	2.9%	1.6%	0.6%	3.2%

Source: Compiled from National Association of College and University Business Officers, *NACUBO-TIAA Study of Endowments* (Washington, National Association of College and University Business Officers, 2019) (and earlier years), an unpublished NACUBO tabulation, and various financial databases.

Note: Fiscal years end on June 30 of each year.
The first average includes endowments that report returns net and gross of fees.
Yale's returns are reported net of fees.
Government bond index is the Barclays U.S. Government/Credit Bond Index.

Exhibit 9 Returns of Yale Endowment, by Asset Class (%)

Asset Class	Target Benchmark (Passive first, Active Below)	Yale 2020 (One Year) Returns	Yale One Year Returns vs. Passive Benchmark	Yale 10-Year Annualized Returns	Yale 10-Year vs. Passive Benchmark	Yale 10-Year vs. Active Benchmark
Domestic Equity	Wilshire 5000	-5.0%	-11.8%	12.8%	-1.0%	0.1%
	BNY Median Manager					
Foreign Developed	MSCI EAFE IMI	9.9%	14.8%	13.6%	7.6%	7.4%
	BNY Median Manager					
Foreign Emerging	Emerging Equity Composite	15.9%	15.6%	18.0%	14.0%	14.2%
	BNY Median Manager					
Fixed Income	Barclays 6-12 Months Treasury	2.3%	-0.2%	1.2%	0.0%	0.0%
	Barclays 6-12 Months Treasury					
Real Estate	MSCI US REIT Index	-2.3%	11.7%	9.7%*	1.0%	-0.8%
	Cambridge Real Estate					
Natural Resources	Natural Resources Composite	-15.6%	24.5%	4.4%*	8.1%	4.0%
	Cambridge Natural Resources					
Venture Capital	Venture Capital Composite	25.0%	15.5%	21.3%*	9.1%	5.6%
	Cambridge Global Venture Capital Composite					
Leveraged Buyouts	Leveraged Buyouts Composite	3.7%	8.4%	14.6%*	4.9%	0.4%
	Cambridge Global Leveraged Buyouts					
Absolute Return	Barclays 9-12 Months Treasury	1.8%	-1.0%	5.3%	4.4%	3.0%
	Credit Suisse Composite					
Total Endowment	Composite Benchmark	6.8%	7.5%	10.9%	4.0%	3.3%
	Cambridge Median Endowment					

Source: Yale University documents.

Note: *Money-weighted returns

Exhibit 10 Returns and Size of Private Equity Investments of the Yale Endowment, 1978-2020

Fiscal Year	Domestic Venture	Foreign Venture	Total Venture	VC Portfolio Value ($M)	Domestic LBO	Foreign LBO	Total LBO	LBO Portfolio Value ($M)	Endowment Value ($M)
1978	27.2%		27.2%	0.8	35.3%		35.3%	2.4	545
1979	-2.2%		-2.2%	1.0	-3.0%		-3.0%	2.4	586
1980	208.1%		208.1%	3.1	231.9%		231.9%	5.3	677
1981	33.3%		33.3%	9.4	-16.6%		-16.6%	6.2	800
1982	25.6%		25.6%	15.9	-47.5%		-47.5%	3.4	748
1983	123.4%		123.4%	31.7	-10.1%		-10.1%	6.8	1,109
1984	3.7%		3.7%	29.1	41.6%		41.6%	8.2	1,071
1985	-10.1%		-10.1%	24.8	5.6%		5.6%	17.2	1,320
1986	2.6%		2.6%	27.0	34.0%		34.0%	20.4	1,751
1987	25.4%		25.4%	38.7	23.9%		23.9%	30.3	2,117
1988	-0.7%	-1.9%	-0.7%	47.5	7.8%		7.8%	36.2	2,056
1989	-0.3%	13.4%	-0.3%	58.2	38.7%		38.7%	46.7	2,333
1990	14.8%	-4.2%	13.3%	79.9	7.3%		7.3%	57.7	2,590
1991	11.4%	-4.7%	9.5%	108.5	14.2%	-30.9%	8.3%	91.1	2,590
1992	28.3%	3.3%	25.4%	154.3	7.2%	7.0%	7.2%	118.7	2,845
1993	13.6%	7.4%	13.0%	183.7	57.3%	-16.5%	53.2%	229.6	3,243
1994	20.2%	24.6%	20.5%	228.4	18.7%	23.1%	18.9%	318.4	3,549
1995	37.8%	12.8%	36.1%	280.3	26.3%	13.3%	25.1%	368.7	3,972
1996	124.8%	10.6%	117.9%	430.1	31.5%	40.4%	32.5%	521.2	4,860
1997	37.6%	17.0%	36.7%	326.2	22.1%	97.9%	37.2%	751.6	5,806
1998	38.0%	47.0%	38.7%	382.6	39.8%	-0.3%	28.5%	878.0	6,592
1999	131.4%	15.5%	131.3%	652.6	8.5%	-16.8%	3.1%	984.4	7,205
2000	701.0%	39.3%	687.8%	1,162.8	35.1%	38.2%	35.6%	1,270.8	10,039
2001	9.0%	-13.4%	8.8%	806.9	-14.7%	-3.5%	-13.4%	1,137.0	10,725
2002	-39.9%	-8.0%	-39.7%	528.4	-11.2%	-0.4%	-9.8%	963.8	10,522
2003	-13.1%	-14.1%	-13.1%	487.8	-0.3%	2.4%	0.1%	1,124.9	11,035
2004	-0.7%	-7.0%	-0.8%	529.0	32.0%	22.7%	30.4%	1,291.6	12,747
2005	25.0%	13.1%	24.8%	598.9	32.4%	19.3%	30.3%	1,596.9	15,224
2006	16.7%	29.4%	17.1%	762.2	35.9%	47.5%	37.7%	2,200.1	18,030
2007	21.7%	41.0%	22.9%	1,042.7	45.1%	57.8%	47.2%	3,152.1	22,530
2008	10.0%	-10.0%	9.9%	1,137.0	-4.6%	21.1%	-0.5%	3,460.9	22,867
2009	-17.2%	-26.2%	-17.3%	1,011.6	-25.9%	-28.8%	-26.5%	2,908.8	16,327
2010	9.7%	-12.4%	9.5%	1,140.8	22.7%	14.8%	21.0%	3,902.9	16,653
2011	39.3%	80.4%	45.4%	2,001.6	22.8%	33.9%	25.2%	4,835.0	19,374
2012	11.9%	-5.9%	8.4%	2,166.1	8.8%	-2.8%	5.9%	4,814.0	19,345
2013	10.9%	-3.1%	8.3%	2,156.5	16.2%	10.3%	14.8%	4,711.6	20,780
2014	44.8%	21.2%	40.8%	3,407.6	19.0%	18.8%	18.9%	4,809.2	23,895
2015	39.6%	33.6%	38.6%	4,364.8	14.9%	4.1%	12.3%	4,349.5	25,572
2016	-3.4%	15.1%	0.0%	4,320.9	8.4%	23.9%	11.4%	3,922.4	25,409
2017	14.4%	1.4%	11.6%	4,877.4	20.2%	28.1%	21.5%	4,066.3	27,176
2018	18.6%	33.1%	21.8%	5,842.0	20.1%	14.3%	19.1%	4,343.3	29,351
2019	26.3%	10.4%	22.1%	6,655.4	12.0%	12.3%	12.1%	5,030.2	30,315
2020	23.5%	27.7%	24.5%	7,554.3	4.1%	2.4%	3.8%	5,290.3	31,202
3-year return	22.6%	23.0%	22.7%		12.0%	9.4%	11.5%		
7-year return	23.0%	19.6%	22.3%		14.3%	14.3%	14.3%		
10-year return	21.9%	18.8%	21.3%		14.9%	13.4%	14.6%		
20-year return	11.0%	18.4%	11.6%		10.9%	12.6%	11.2%		
Since Inception	34.5%	15.8%	33.7%		17.0%	15.8%	16.8%		

Source: Yale University documents. Returns are for year ending June 30 of each year. Value of private equity portfolio and endowment are as of June 30 and are expressed in millions of dollars. The Yale fiscal year returns are internal rates of return calculated on a daily basis. Multiyear returns are based on internal rates of return using quarterly data.

Endnotes

[1] This section is based on Brooks Mather Kelley, *Yale: A History* (New Haven: Yale University Press, 1974); David F. Swensen, *Pioneering Portfolio Management: An Unconventional Approach to Investment Management* (New York: Free Press, 2000); and Yale University Investments Office, *The Yale Endowment* (New Haven: Yale University, 2006).

[2] General university information is from Institutional Department, Scudder, Stevens & Clark, *Survey of University and College Endowment Funds* (New York: Scudder, Stevens & Clark, 1947).

[3] Advisory Committee on Endowment Management, *Managing Educational Endowments: Report to the Ford Foundation* (New York: Ford Foundation, 1969).

[4] Yale News "Investment return of 6.8% brings Yale endowment value to $31.2 billion", September 24, 2020 https://news.yale.edu/2020/09/24/investment-return-68-brings-yale-endowment-value-312-billion, accessed November 24, 2020

[5] R. Alexander Hetherington, cited in Griswold, "Investments Office Hires Yalies."

[6] John M. Keynes, *The General Theory of Employment Interest and Money* (New York: Harcourt Brace, 1936), Chapter 12.

[7] Data from *Ibbotson "SBBI" 2019* (Chicago, Illinois: Morningstar, 2019), accessed December 3, 2020

[8] Yale actively rebalanced its portfolio to maintain its target asset allocations, however, and this led to frequent short-term adjustments in its holdings. For instance, as equity values rose in the summer of 1987, Yale sold stocks in order to return to its target allocation level. After the stock market crash later that year, the endowment repurchased many of the same securities as it sought to bring its asset allocation back to the target level.

[9] Cambridge Associates data, https://www.sec.gov/files/cambridge-associates-private-investments.pdf, accessed November 30, 2020

[10] David F. Swensen, Dean J. Takahashi, and Timothy R. Sullivan, "Private Equity—Portfolio Review," memorandum to the Investment Committee, September 20, 2010, p. 2.

[11] Yale News "Investment return of 6/8% brings Yale endowment value to $31.2 billion", September 24, 2020 https://news.yale.edu/2020/09/24/investment-return-68-brings-yale-endowment-value-312-billion, accessed November 24, 2020

[12] Cambridge Associates data for endowments with assets over $1 billion.

[13] The target spending rate currently stands at 5.25%. According to the smoothing rule, the amount of the endowment spent in a given year sums to 80% of the previous year's spending and 20% of the targeted long-term spending rate applied to the fiscal year-end market value two years prior. The spending amount determined by the formula is adjusted for inflation and an allowance for taxes and constrained so that the calculated rate is at least 4.0%, and not more than 6.5%, of the endowment's inflation-adjusted market value two years prior. The smoothing rule and the diversified nature of the Endowment are designed to mitigate the impact of short-term market volatility on the flow of funds to support Yale's operations.

[14] David F. Swensen, Dean J. Takahashi, and Timothy R. Sullivan, "Private Equity—Portfolio Review," memorandum to the Investment Committee, September 29, 1994, p. 5.

[15] David F. Swensen, Dean J. Takahashi, and Timothy R. Sullivan, "Private Equity—Portfolio Review," memorandum to the Investment Committee, September 30, 2010, p. 2.

[16] David F. Swensen, Dean J. Takahashi, Timothy R. Sullivan, Alan S. Forman, and Seth D. Alexander, "Private Equity—Portfolio Review," memorandum to the Investment Committee, October 7, 1999, p. 15.

[17] David F. Swensen, Dean J. Takahashi, and Timothy R. Sullivan, "Private Equity—Venture Capital Strategy," memorandum to the Investment Committee, March 4, 1992, p. 7.

[18] When Sequoia instituted a rule that LPs could only invest in its main U.S. early stage fund if they also invested in its China, India, or growth funds, Yale declined to participate, despite a long-standing association with the firm.

[19] This was in contrast to venture or buyout investing, where individual partners' successes and failures could be more or less assessed by examining who represented the partnership as a director on various firms' boards.

[20] For more on the model, see Dean Takahashi and Seth Alexander, "Illiquid Alternative Asset Fund Modeling," *Journal of Portfolio Management* 28, no. 2 (Winter 2002).

[21] Levy, A. "A low-profile investor who bet on Snowflake 8 years ago is up more than $12B after the IPO pop" CNBC News, September 16, 2020 https://www.cnbc.com/2020/09/16/snowflake-investor-sutter-hill-holds-11point6-billion-after-ipo-pop.html

[22] For example, valuation issues arise in terms of the estimates used in the spending rule, which originally assumed that market prices would be available to value the assets.